Topic 7: Schedule Trigger Automation

> **Institution:** Tayana Academy | LMS Assessment
> **Topic:** N8N Hands-On Beginner | Topic 7: Schedule Trigger Automation
> **Workflow Name:** `Topic7_Schedule_Trigger_Assessment`
> **Target Audience:** Trainee / Employee
> **Difficulty Level:** Beginner
> **Platform Version:** n8n version 1.0.0 or higher
> **Student:** Abhishek Nikam (`abhishek.n@tayanasolutions.com`)
> **Repository:** n8n-assessments

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1. Assessment Overview & Objectives

This practical assessment evaluates the learner's capability to design, configure, test, activate, and document an automated time-driven workflow in n8n using the **Schedule Trigger** node. Moving beyond manual testing triggers, this workflow establishes two distinct scheduling rules within a single trigger node: a daily execution rule at 09:00 AM and a custom weekday cron rule at 08:30 AM (`30 8 * * 1-5`), logging dynamic runtime execution timestamps into an downstream action node.

Key Deliverables in this Package:

1. `README.md` : Comprehensive assessment documentation covering architecture, cron syntax breakdown, dynamic timestamp logging, activation lifecycle, and LMS submission notes.
2. `Topic7_Schedule_Trigger_Assessment.json` : Official exportable n8n workflow file pre-configured with active status and dual schedule rules.
3. `README.pdf` : Print-ready, professionally styled PDF document generated directly from this documentation.

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2. Workflow Architecture & Scheduling Pipeline

The automation runs autonomously based on background schedule events:

```
flowchart LR
    subgraph TriggerNode["⏰ Schedule Trigger Node (Multi-Rule)"]
        R1["Rule 1: Every Day @ 09:00 AM"]
        R2["Rule 2: Custom Cron '30 8 * * 1-5' (Mon-Fri @ 08:30 AM)"]
    end

    TriggerNode --> B["📝 Log Execution<br/>(Capture Dynamic Timestamp)"]

    subgraph OutputData["Logged Output Payload"]
        B1["executionTime: 'YYYY-MM-DD HH:mm:ss' (Dynamic)"]
    end

    B -.-> OutputData

    classDef trigger fill:#10b981,stroke:#047857,stroke-width:2px,color:#fff;
    classDef set fill:#6366f1,stroke:#4338ca,stroke-width:2px,color:#fff;
    class TriggerNode trigger;
    class B set;
```

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3. Node Specifications & Detailed Configuration

Node 1: Schedule Trigger

- **Node Label:** `Schedule Trigger`
- **Node Type:** `n8n-nodes-base.scheduleTrigger` (typeVersion: `1.2`)
- **Canvas Check:** Contains **no** Manual Trigger node; workflow initiation is purely time-governed.
- **Configured Schedule Rules:**

Rule #	Schedule Type	Configuration Parameters	Natural Language Trigger Schedule
Rule 1	Daily Interval (`days`)	`triggerAtHour: 9`, `triggerAtMinute: 0`	Every day at 09:00 AM
Rule 2	Custom Cron (`cronExpression`)	`30 8 * * 1-5`	Every weekday (Monday to Friday) at 08:30 AM

Node 2: Log Execution

- **Node Label:** `Log Execution`
- **Node Type:** `n8n-nodes-base.set` (typeVersion: `3.4`)
- **Fields to Set:**

Field Name	Data Type	Dynamic Expression Syntax	Evaluated Result
`executionTime`	`String`	`={{ $now.toFormat('yyyy-MM-dd HH:mm:ss') }}`	Current execution timestamp (e.g. `"2026-10-13 09:00:00"`)

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4. Deep Dive: 5-Field Standard Cron Syntax Breakdown

Rule 2 configures the custom cron expression: `30 8 * * 1-5`

```
┌──────────── minute (0 - 59) -> 30 (30th minute of the hour)
│ ┌────────── hour (0 - 23)   -> 8 (8 AM in 24-hour time)
│ │ ┌──────── day of month    -> * (every day of the month)
│ │ │ ┌────── month (1 - 12)  -> * (every month of the year)
│ │ │ │ ┌──── day of week     -> 1-5 (Monday through Friday, 1=Mon ... 5=Fri)
│ │ │ │ │
30 8 * * 1-5
```

Field-by-Field Analysis:

1. `30` **(Minute):** Specifies execution at minute 30 past the hour.
2. `8` **(Hour):** Specifies hour 8 (08:00 AM). Combined with minute 30, this targets 08:30 AM.
3. `*` **(Day of Month):** Asterisk wildcard matches every calendar day (1 through 31).
4. `*` **(Month):** Asterisk wildcard matches every month (January through December).
5. `1-5` **(Day of Week):** Hyphenated range matching days 1 (Monday) through 5 (Friday). Weekends (Saturday=6, Sunday=0/7) are completely excluded.

- **Combined Translation:** *"At 08:30 AM, Monday through Friday."*

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5. Assessment Tasks & Step-by-Step Walkthrough

Exercise 1 – Create and Name the Scheduled Workflow

- Created a fresh blank workflow canvas.
- Renamed the workflow exactly to:

```
Topic7_Schedule_Trigger_Assessment
```

- Placed the `Schedule Trigger` node as the starting entry node.
- Confirmed that the canvas contains **no Manual Trigger node**.

Exercise 2 – Configure the Daily Schedule Rule

- Opened the `Schedule Trigger` configuration panel.
- Under **Trigger Interval**, added the first rule:
  - Interval Mode: `Days` / `Every Day`
  - Trigger at Hour: `9`
  - Trigger at Minute: `0`
- Confirmed evaluation matches daily execution at 09:00 AM.

Exercise 3 – Configure the Custom Weekday Cron Rule

- Within the **same** Schedule Trigger node, added a second trigger rule item.
- Selected schedule mode: **Custom (Cron)**.
- Entered exact cron expression:

```
30 8 * * 1-5
```

- Validated that both rules coexist concurrently in the same trigger node.

Exercise 4 – Capture and Verify the Execution Timestamp

- Connected `Schedule Trigger` to an Edit Fields (Set) node renamed to **`Log Execution`**.
- Created field `executionTime` of type **String**.
- Configured dynamic expression:

```
{{ $now.toFormat('yyyy-MM-dd HH:mm:ss') }}
```

- Executed node test via **Test step** / manual trigger simulation.
- Verified that output contains a real-time ISO/formatted date string rather than static text.

Exercise 5 – Activate, Document, and Submit the Workflow

- Saved the workflow (`Ctrl+S` / `Cmd+S`).
- Toggled the workflow status switch in the top-right header to **Active** (`active: true`).
- Exported the completed workflow as `Topic7_Schedule_Trigger_Assessment.json`.
- Generated publication PDF `README.pdf`.
- Updated centralized public repository index at `3OOM/n8n-assessments`.

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6. Output Verification & Multi-View Inspection

Final Verified JSON Output Payload

```
[
  {
    "executionTime": "2026-10-13 09:00:00"
  }
]
```

Multi-View Inspection Matrix

View Mode	What Evaluator Observes	Verification Criteria Met
Table View	Single column `executionTime` displaying formatted datetime string (e.g. `2026-10-13 09:00:00`).	Clean tabular presentation of runtime log.
JSON View	Clean key-value JSON structure with property `executionTime` reflecting dynamic timestamp.	Exact match: no static/mock string literals used.
Schema View	• `executionTime` : `string`	Confirms strict primitive string type adherence.

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7. Practical Assessment Checklist

- [x] Workflow created and named **`Topic7_Schedule_Trigger_Assessment`**.
- [x] Schedule Trigger node configured as starting entry node (no Manual Trigger on canvas).
- [x] Daily schedule rule configured for 09:00 AM.
- [x] Second schedule rule added in the same trigger node.
- [x] Custom cron expression `30 8 * * 1-5` configured correctly.
- [x] 5-field cron syntax accurately explained in documentation.
- [x] `Log Execution` node connected and renamed.
- [x] `executionTime` field created as String.
- [x] Dynamic `$now` expression implemented (`{{ $now.toFormat('yyyy-MM-dd HH:mm:ss') }}`).
- [x] Workflow test completed successfully with green indicators.
- [x] Dynamic timestamp output verified across Table, JSON, and Schema views.
- [x] Workflow saved and activated (`active: true`).
- [x] Workflow JSON exported (`Topic7_Schedule_Trigger_Assessment.json`).

- [x] Print-ready PDF generated (`README.pdf`).
- [x] Pushed all deliverables to public GitHub repository.

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8. LMS Submission Description & Assessor Notes

> **Copy and paste the text below directly into the LMS Description field when submitting:**

```
================================================================================
TAYANA ACADEMY LMS SUBMISSION | TOPIC 7: SCHEDULE TRIGGER AUTOMATION
================================================================================
Student Name: Abhishek Nikam
Student Email: abhishek.n@tayanasolutions.com
Workflow Name: Topic7_Schedule_Trigger_Assessment
GitHub Repository: https://github.com/3OOM/n8n-assessments/tree/main/topic-07-schedule-trigger

1. WORKFLOW IMPLEMENTATION SUMMARY:
- Constructed an automated time-triggered pipeline: Schedule Trigger -> Edit Fields (Log Execu
- Canvas starts strictly with Schedule Trigger (no Manual Trigger present).
- Configured dual schedule rules in the single Schedule Trigger node:
  * Rule 1 (Daily): Every day at 09:00 AM (hour: 9, minute: 0).
  * Rule 2 (Custom Cron): 30 8 * * 1-5 (At 08:30 AM, Monday through Friday).
- "Log Execution" records runtime audit info:
  * executionTime (String): {{ $now.toFormat('yyyy-MM-dd HH:mm:ss') }}
- Saved and set workflow to "Active" state for background daemon scheduler evaluation.

2. CRON SYNTAX BREAKDOWN (30 8 * * 1-5):
- Minute: 30 (30th minute)
- Hour: 8 (08:00 AM in 24-hour notation)
- Day of Month: * (every calendar day)
- Month: * (every month)
- Day of Week: 1-5 (Monday to Friday, excluding Saturday and Sunday)
- Operational Meaning: Triggers automatically at 8:30 AM every weekday.

3. TIMEZONE CONSIDERATIONS & RUNTIME ENVIRONMENT:
- n8n evaluates cron schedules against the instance timezone (configured via GENERIC_TIMEZONE
- Dynamic formatting using $now.toFormat('yyyy-MM-dd HH:mm:ss') captures the precise local run

4. VERIFICATION ACROSS MULTI-VIEW MODES:
- Table View: Verified executionTime column renders clean timestamp string.
- JSON View: Confirmed payload strictly contains executionTime without static hardcoded text.
- Schema View: Confirmed string primitive type preservation.
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```